that the requested absorption is achieved.

Objective evaluated on physical predictions

For a predicted geometry, report minimum and average simulated absorption over the requested band, and the nonnegative threshold violation. Prefer minimum in-band absorption as the principal physical success metric because it detects narrow absorption gaps hidden by averages.

$$V(q,\hat{g}) = \max\left(0, X - \min_{\lambda_i \in B} A_{\mathrm{HFSS}}(\lambda_i;\hat{g})\right)$$

The reported physical metric must come from an independent HFSS verification solve. Neural classification confidence is not a calibrated probability that the target is physically feasible.

5. Inference, evaluation and limits

At inference, no population, global optimizer, gradient-based geometry search, neural surrogate or second model is used. The one trained MLP provides one geometry directly.

Stage	Operation
1. Request	Receive a, b, X; accept any 450 ≤ a < b ≤ 1600 nm, 0 ≤ X ≤ 1
2. Inference	Normalize (a, b, X); execute exactly one MLP forward pass
3. Decode	Argmax over eight classes; decode hard-sigmoid coordinates, including 0 and 320 nm
4. Return	Report shape, dimensions, substrate, operating assumptions and a feasibility disclaimer
5. Verification	Run HFSS once for the returned geometry when physical correctness matters; verification is not a search

Deliverables and performance measures

- Categorical accuracy: family + substrate, against the deterministic catalogue label.
- Dimensional error: outer/inner geometric parameter error in nm on held-out examples.
- Physical threshold violation V, using full-wave HFSS predictions instead of neural estimates.
- Feasible-request success rate and mean minimum absorption over the requested band.
- Generalization disclosure: normal-incidence TM only; no claims about TE or oblique incidence.

Limits imposed by the chosen restrictions

The baseline contains only 80 geometric parameter settings across the full 0–320 nm domain. Very small gaps, walls and narrow-band resonances are therefore underrepresented or numerically unresolved. Artificially generated query/label pairs do not replace additional electromagnetic diversity. With one direct inverse network and no tuning or runtime search, report observed accuracy and the HFSS verification status instead of asserting that the model finds a globally best geometry.

Two-week dataset connection

The second document uses the same fixed 160 standard HFSS simulations and 8 higher-fidelity checks as Document 01. The 14-day capacity assumption applies only to data generation. Neural training, new held-out verification solves, and potential retries must be scheduled separately unless covered by the reserved verification budget.

References and provenance

[1] Dataset and Problem Formulation, companion PDF, sections 2–6.

[2] Liu et al., Training Deep Neural Networks for the Inverse Design of Nanophotonic Structures. ACS Photonics 2018. https://doi.org/10.1021/acsphotonics.7b01377

[3] Ivan.pdf, Batende.pdf, Yaislin.pdf and repository docs/problem.pdf. GitHub and Ansys source links are in the companion PDF.